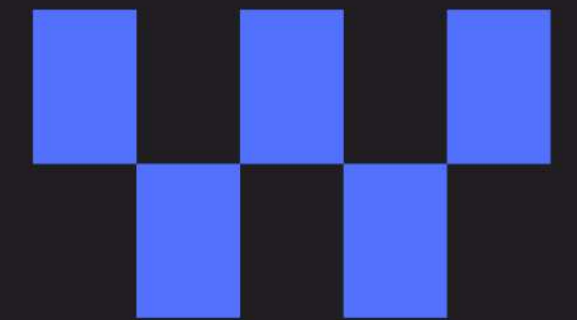


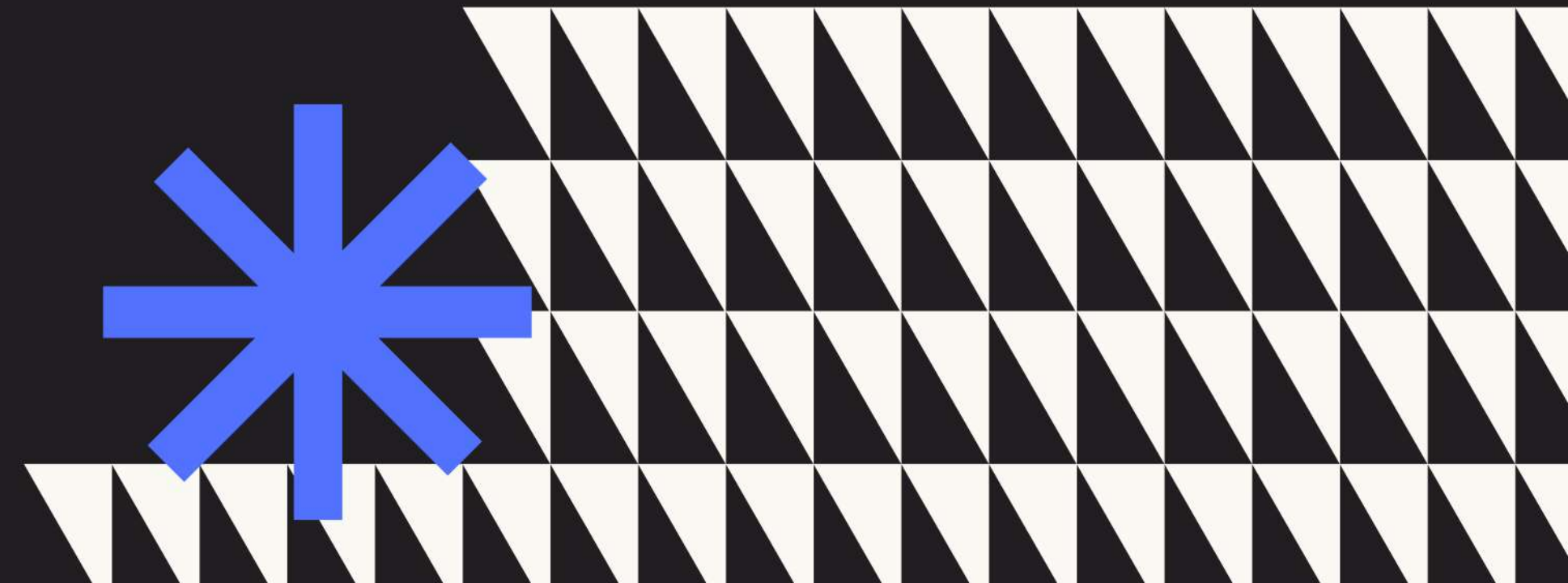
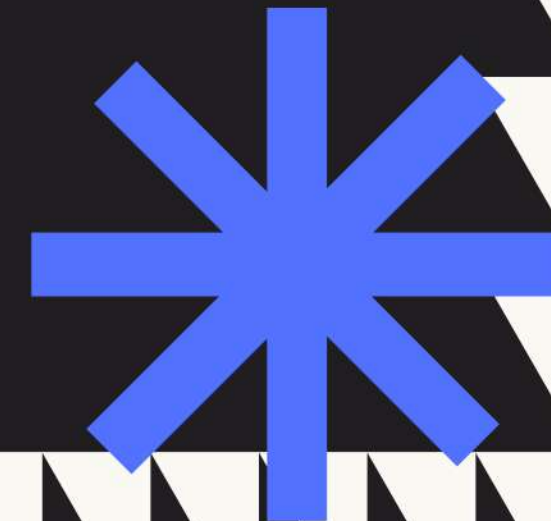
Inter IIT TechMeet

Team – Akelahu

BootCamp – Strategy



Nikunj Sharma
Mentor -- Anirban Ghosh



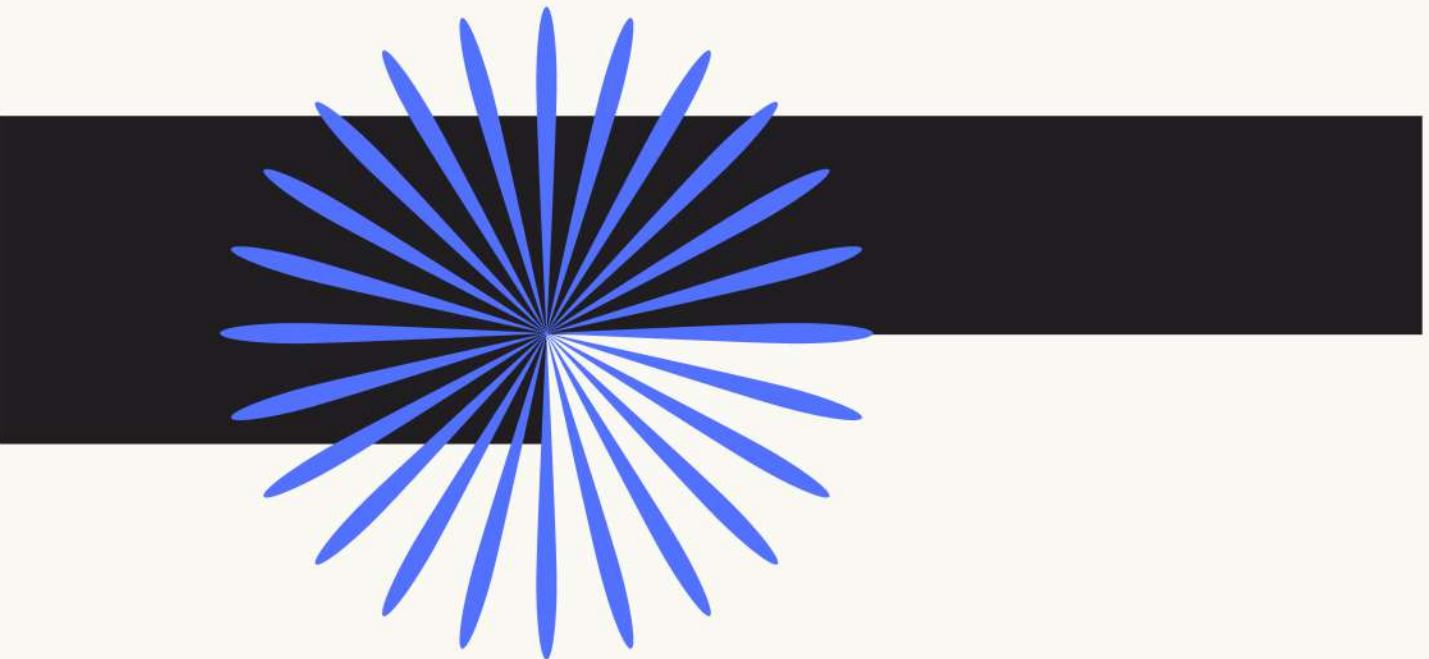
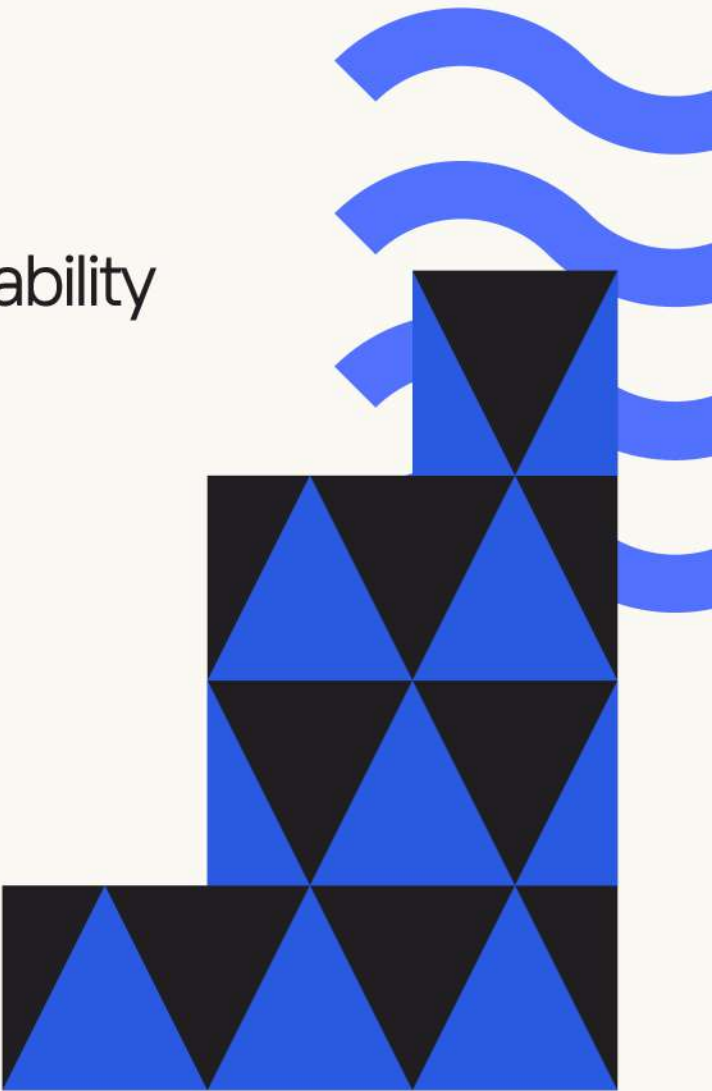


Table of Content

- 01** Introduction
- 02** Current State of Agricultural Emissions
- 03** Agricultural Supply Chain Analysis
- 04** Emission Hotspots in the Supply Chain
- 05** Data Analysis and Results
- 06** Strategies for Emission Reduction

- 07** Supply Chain Workflow Diagram
- 08** Balancing Carbon Emissions and Profitability
- 10** Prediction
- 11** Implementation Strategy
- 12** Impact Assessment & Risk Analysis
- 13** Conclusions and Recommendations



Introduction

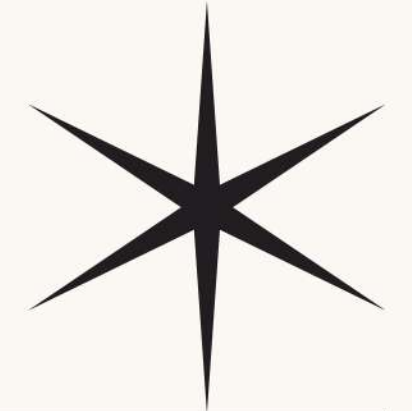
- **Background on Agricultural Carbon Emissions:** Agriculture significantly contributes to global carbon emissions, with activities like fertilizer application, livestock management, and land use changes being key contributors. Understanding these emissions is crucial for developing effective mitigation strategies.
- **Importance of Addressing Emissions in Agriculture:** Reducing agricultural carbon emissions is vital for combating climate change, enhancing food security, and promoting sustainable farming practices. Effective emission management can lead to improved soil health and biodiversity.

Scope of the Report:

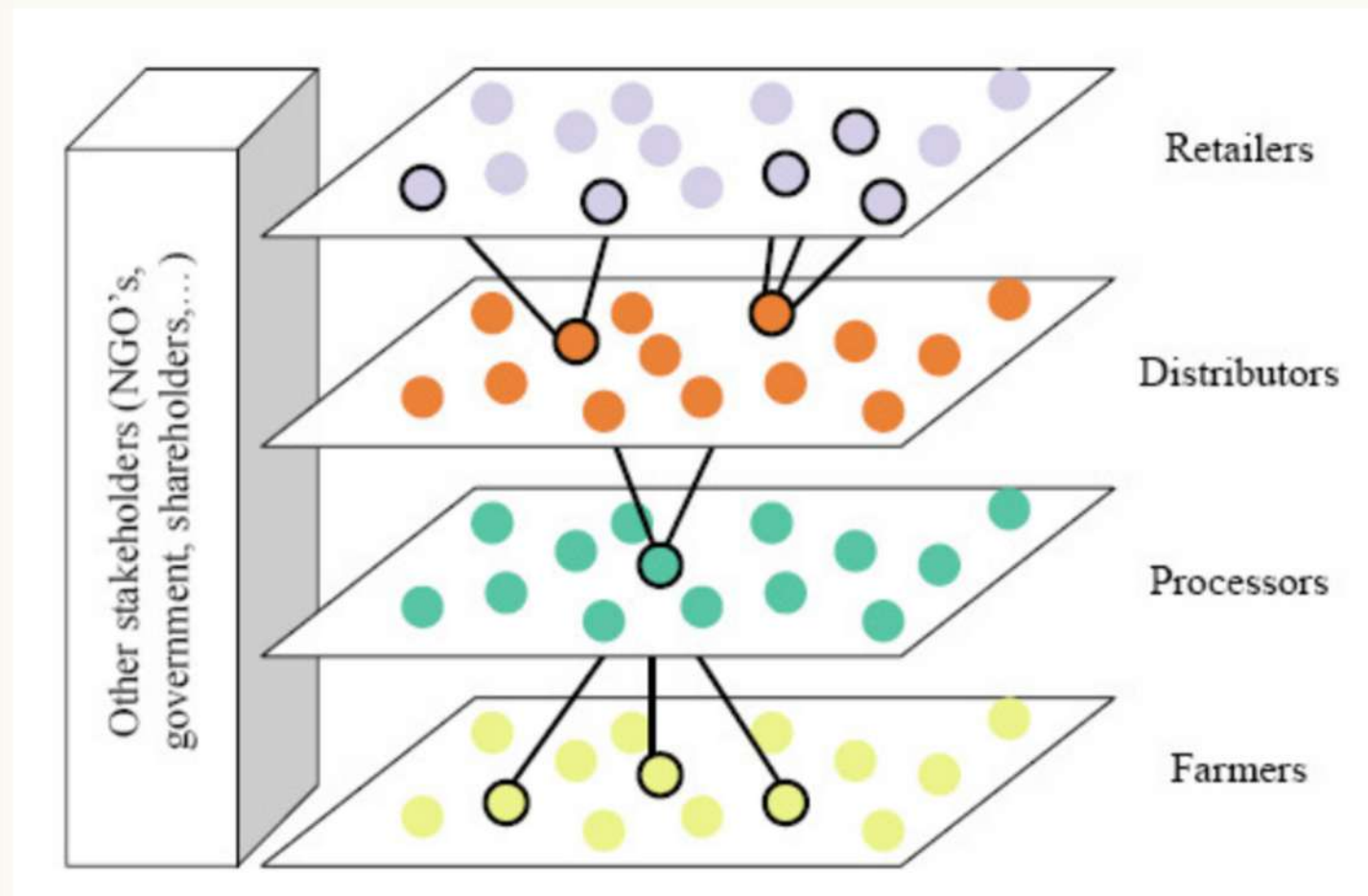
- Analyze current agricultural practices and their impact on carbon emissions.
- Explore sustainable alternatives and technologies.
- Provide actionable recommendations for stakeholders in the agriculture supply chain.



Current State of Agricultural Emissions

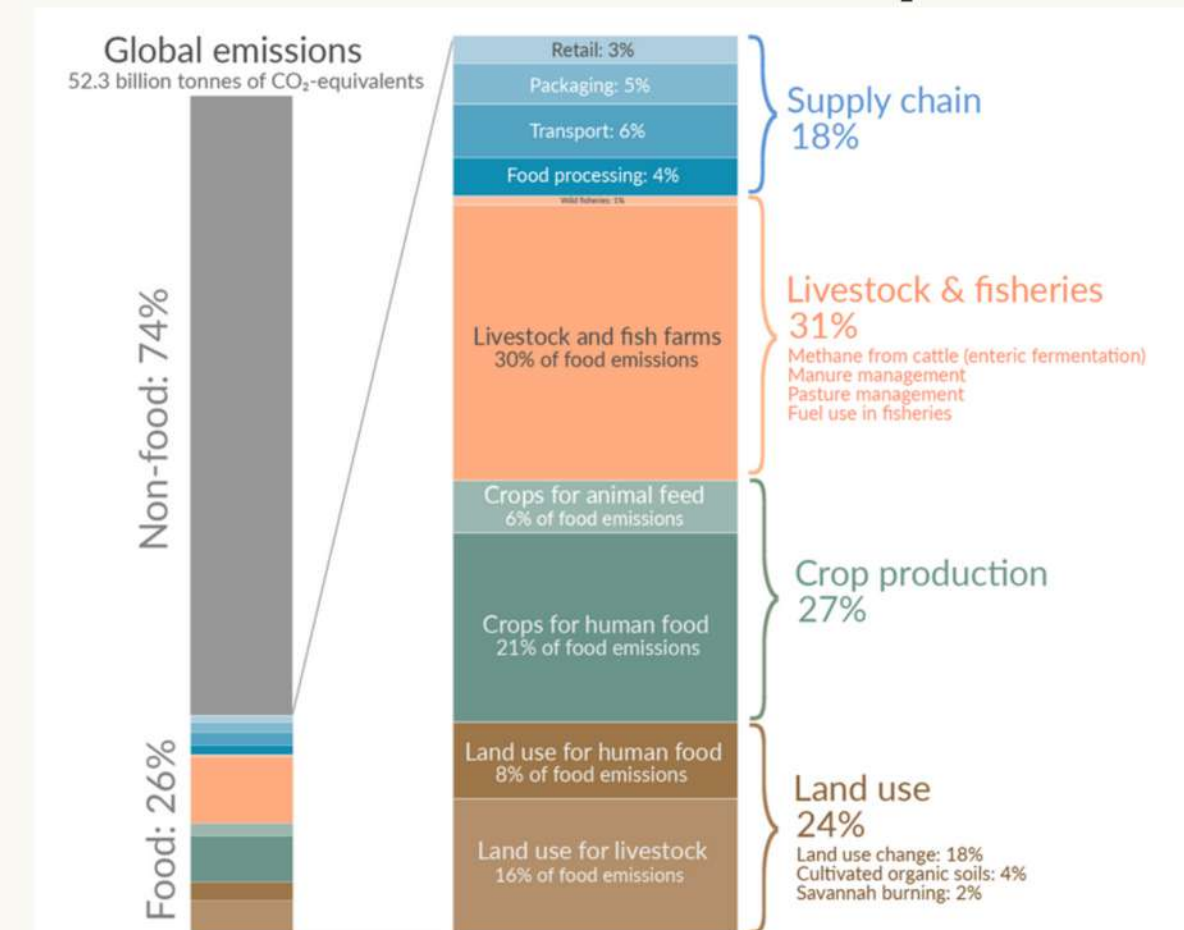


Schematic Diagram of Supply Chain



The typical agriculture supply chain involves three steps: from farmers to intermediate silos, from silos to transformation plants, and from transformation plants to clients.

Environmental Impact

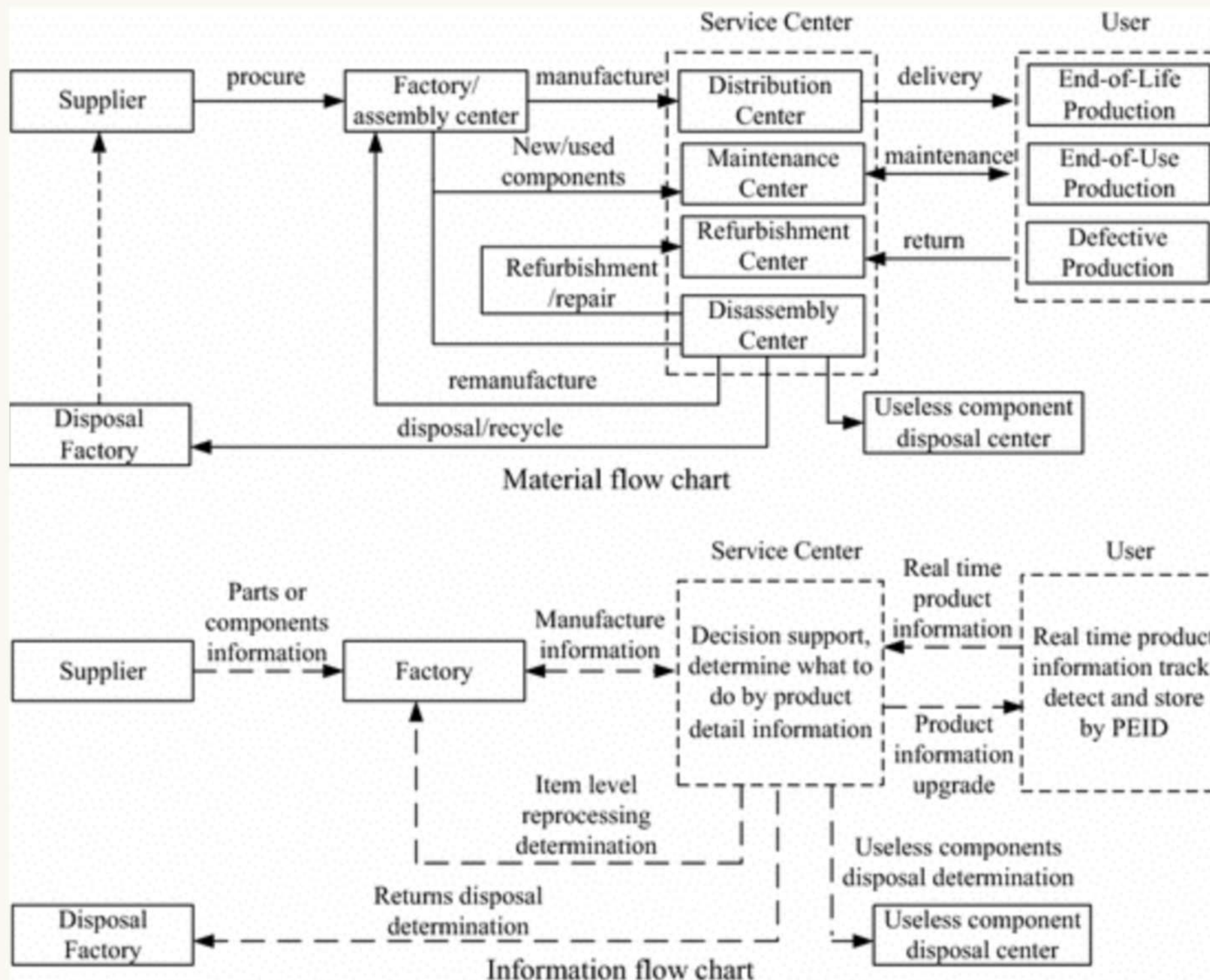


Agriculture is responsible for approximately 26% of total greenhouse gas emissions globally, In which **supply chain is 18%** significantly impacting climate change.

To maximize value from agriculture supply chains, players must make multiple decisions

Agricultural Supply Chain Analysis

Overview of the Agricultural Supply Chain



Agricultural supply chains are responsible for approximately one-fourth of global greenhouse gas emissions. Recent studies indicate that implementing sustainable practices across the supply chain could **reduce these emissions by up to 50%**. Key areas for improvement include precision agriculture, which can **reduce fertilizer use by 15-20%**, and **optimized transportation logistics**, potentially **cutting fuel consumption by 25%**. Additionally, **reducing food waste**, which accounts for 8% of global emissions, is crucial. **Blockchain technology** for traceability and **AI-driven predictive analytics** are emerging as powerful tools, with early adopters reporting up to 30% improvements in supply chain efficiency. As climate change threatens crop yields, resilient and sustainable agricultural supply chains are becoming not just environmentally necessary, but economically imperative for global food security.

Emission Hotspots in the Supply Chain

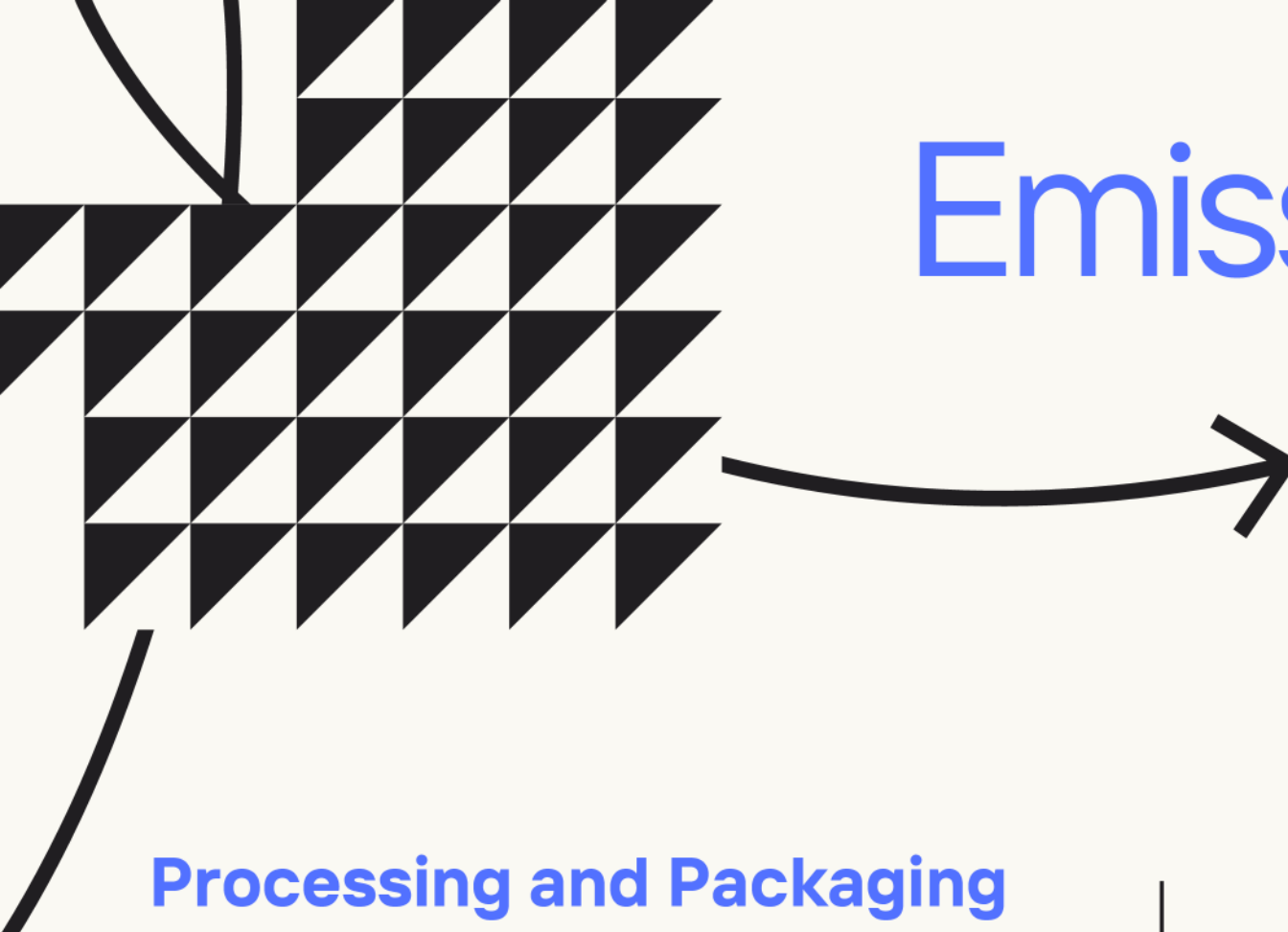
Farm-Level Emissions

Farm operations represent a **major emission source**, with a focus on activities like land use changes, **fertilizer** application, and **methane emissions** from livestock.

Land-use changes, particularly deforestation for agriculture, lead to substantial carbon emissions. Fertilizers (eg **nitrogen based**) produce nitrous oxide, a potent greenhouse gas, while livestock generate methane through enteric fermentation and manure handling. Precision agriculture and regenerative farming practices offer significant opportunities to mitigate these emissions.

EMISSIONS IN FOOD SUPPLY CHAIN





Emission Hotspots in the Supply Chain

Processing and Packaging

Energy consumption in agricultural processing is an emission-heavy stage, with energy used for drying, milling, or refining. The emissions extend into packaging, where materials such as plastics or metals increase the carbon footprint. Companies have begun experimenting with biodegradable packaging or lightweighting packaging materials to reduce these emissions.

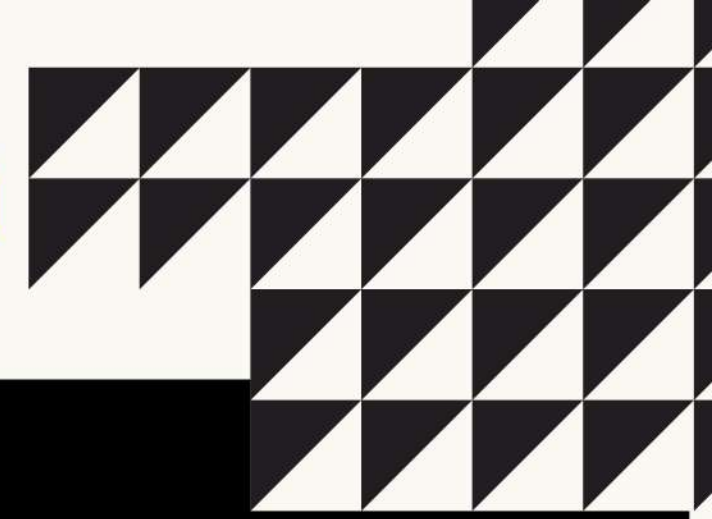
Transportation and Distribution

Agricultural products often travel long distances from farms to processing facilities, and then to retail markets, making transportation a key emission contributor. Transportation emissions are compounded by inefficient logistics, long supply routes, and the reliance on fossil fuels. The trend towards local sourcing and improving supply chain visibility through digital tools (e.g., IoT, GPS tracking) can help reduce carbon footprints in this stage.

Retail and Consumption

Retailers contribute emissions through energy-intensive refrigeration, storage, and operational processes. Additionally, consumer behavior can drive indirect emissions through food waste and inefficient household energy use (e.g., inefficient cooking or refrigeration appliances). Reducing food waste through supply chain transparency and improving consumer education on sustainability can mitigate these emissions.

Emission Hotspots in the Supply Chain



Global agrifood system GHG emissions by lifecycle stage (billion tons CO₂eq) and per capita emissions (tons CO₂eq), 1990-2019

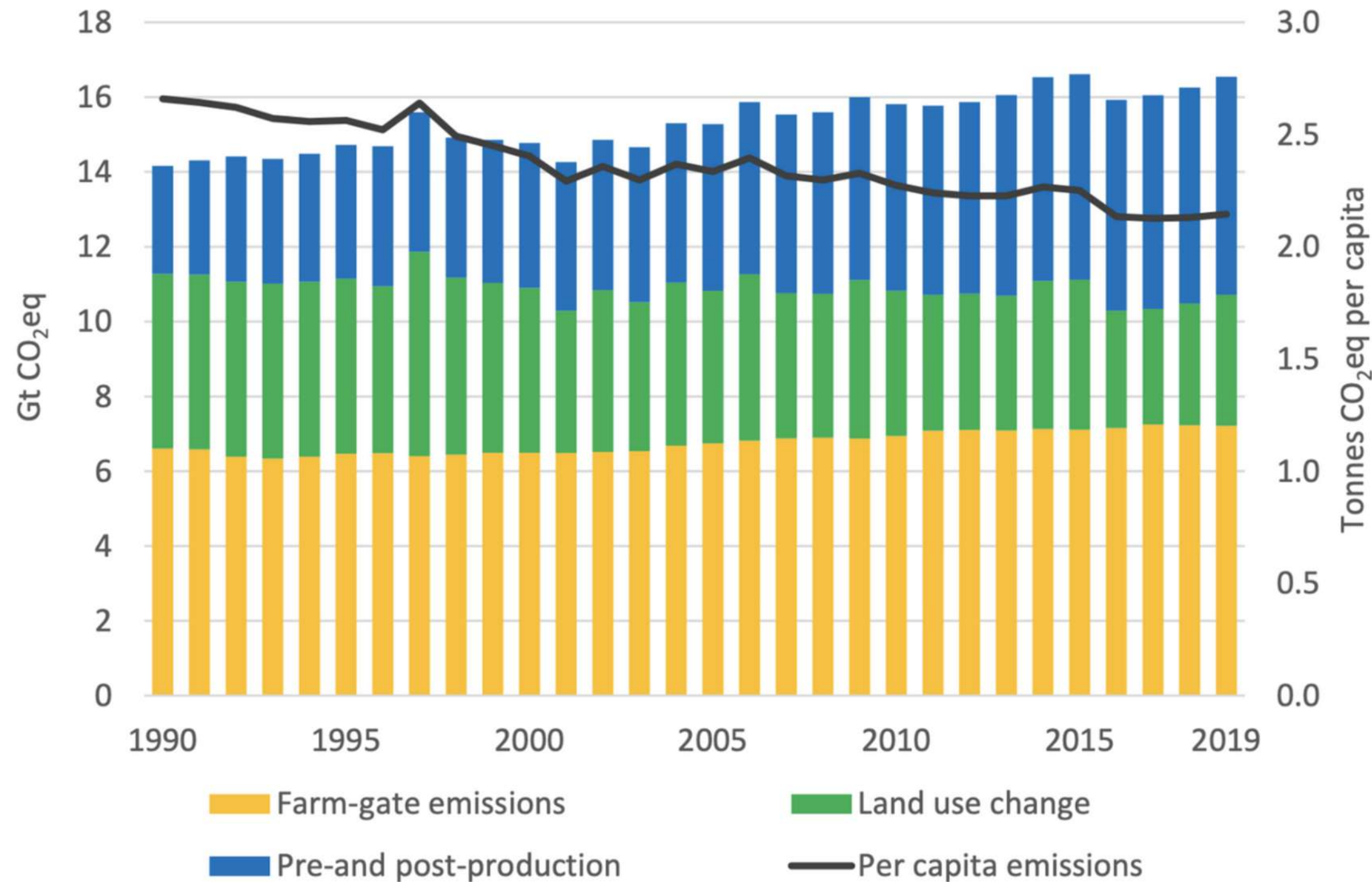


Image credit: FAO

Identification of Key Emission Hotspots and Inefficiencies

Retail and Consumption

Retailers contribute emissions through **energy-intensive refrigeration**, storage, and operational processes. Additionally, consumer behavior can drive indirect emissions through food waste and inefficient household energy use (e.g., inefficient cooking or refrigeration appliances). Reducing food waste through **supply chain transparency** and improving **consumer education** on sustainability can mitigate these emissions.

Data Analysis and Results

Quantitative Analysis of Supply Chain Emissions

Descriptive Statistics

Globally, agriculture is responsible for **26% of total greenhouse gas (GHG)** emissions, with **livestock alone contributing 14.5%**. Emission intensity varies significantly across agricultural sectors, averaging **1.6 kg CO₂** equivalent per kg of food. For example, **rice cultivation releases high levels of methane**, while horticulture has a lower carbon footprint due to more controlled environments.

Inferential Statistics

Statistical analysis revealed that larger farms tend to have lower emissions per unit of output due to economies of scale. **Organic farming reduces emissions by 10-20%** compared to conventional farming, particularly regarding nitrogen-based fertilizers. Regression models identified key relationships between emissions, farm size, and the use of sustainable practices.

Modeling and Simulations

Simulations demonstrated that improvements in logistics, such as **optimizing transport routes, could reduce total emissions by 15%**. Transitioning to **electric vehicles for transport was found to cut transportation-related emissions by 30-50%**, while reducing food waste through better packaging could decrease emissions from the distribution phase.

Data Analysis and Results

Qualitative Analysis of Supply Chain Practices

Thematic Analysis of Interviews

Key insights from interviews with stakeholders across the supply chain included **barriers to adopting sustainable practices**, such as the high upfront costs of low-emission technologies. The lack of policy incentives for sustainable farming was identified as a significant challenge .

Case Study Syntheses

A case study in India showed that the adoption of **precision agriculture and drip irrigation resulted in a 40% reduction in water and energy consumption**. In contrast, regions dependent on conventional farming methods experienced higher emissions. This highlights the importance of technology adoption and government support in driving sustainability .

Comparative Analysis Across Different Agricultural Sub-sectors

Across different sub-sectors, livestock farming contributes significantly more emissions compared to plant-based agriculture. **Livestock emits 4-5 times more GHG per kg of protein**. Rice farming, particularly in Asia, is a major contributor to methane emissions, whereas horticulture (especially in controlled environments) demonstrates a smaller carbon footprint .

Strategies for Emission Reduction

Digital Twin in Supply Chain

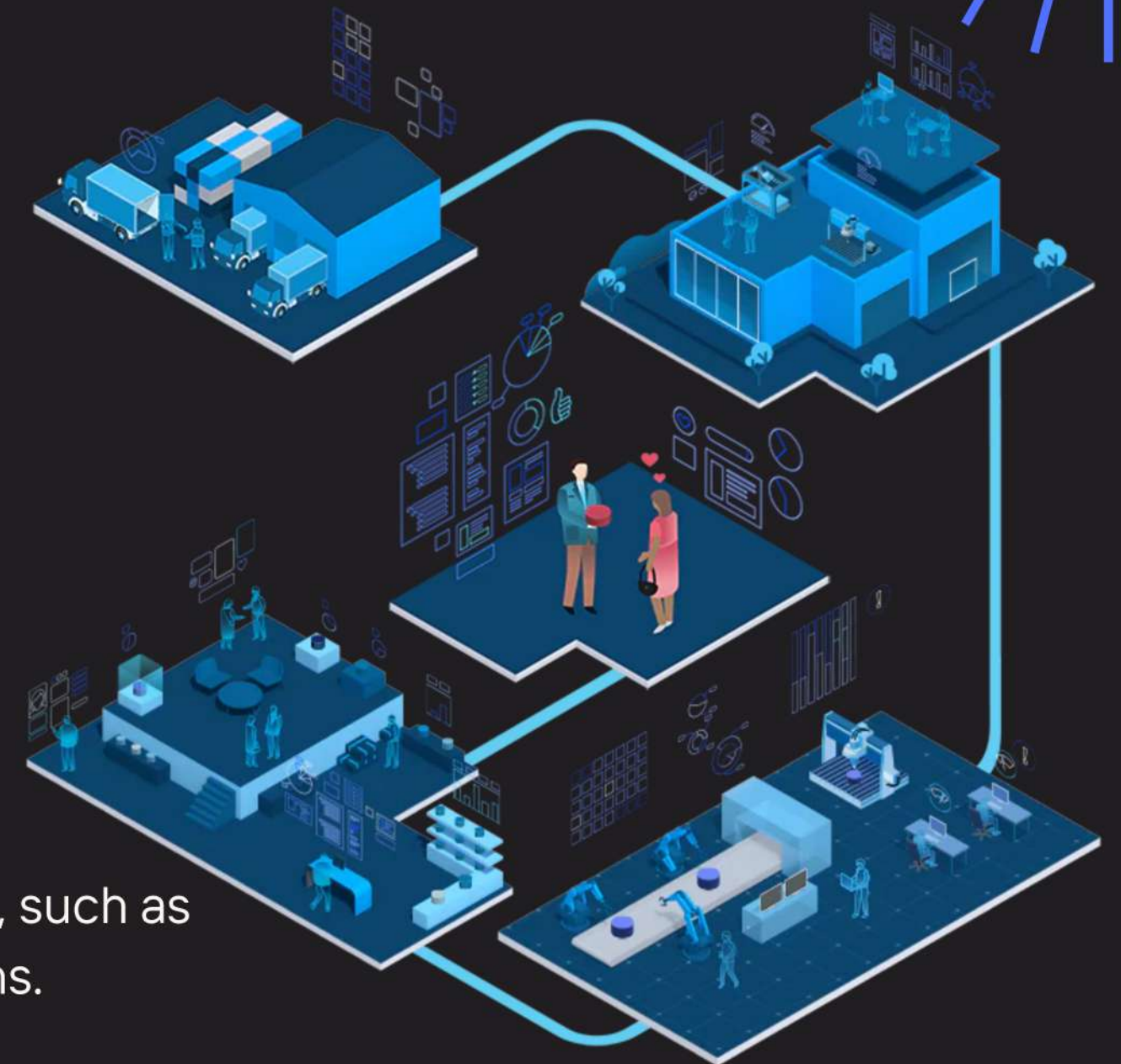
A digital twin is a virtual model of the supply chain that integrates real-time data to simulate and predict supply chain behavior. By **replicating the physical world** digitally, companies can identify inefficiencies, optimize processes, and anticipate potential disruptions. For example, it can predict delays in transportation or processing, allowing for proactive adjustments, which in turn reduces carbon emissions by improving logistics, resource use, and overall operational efficiency.

Blockchain Technology

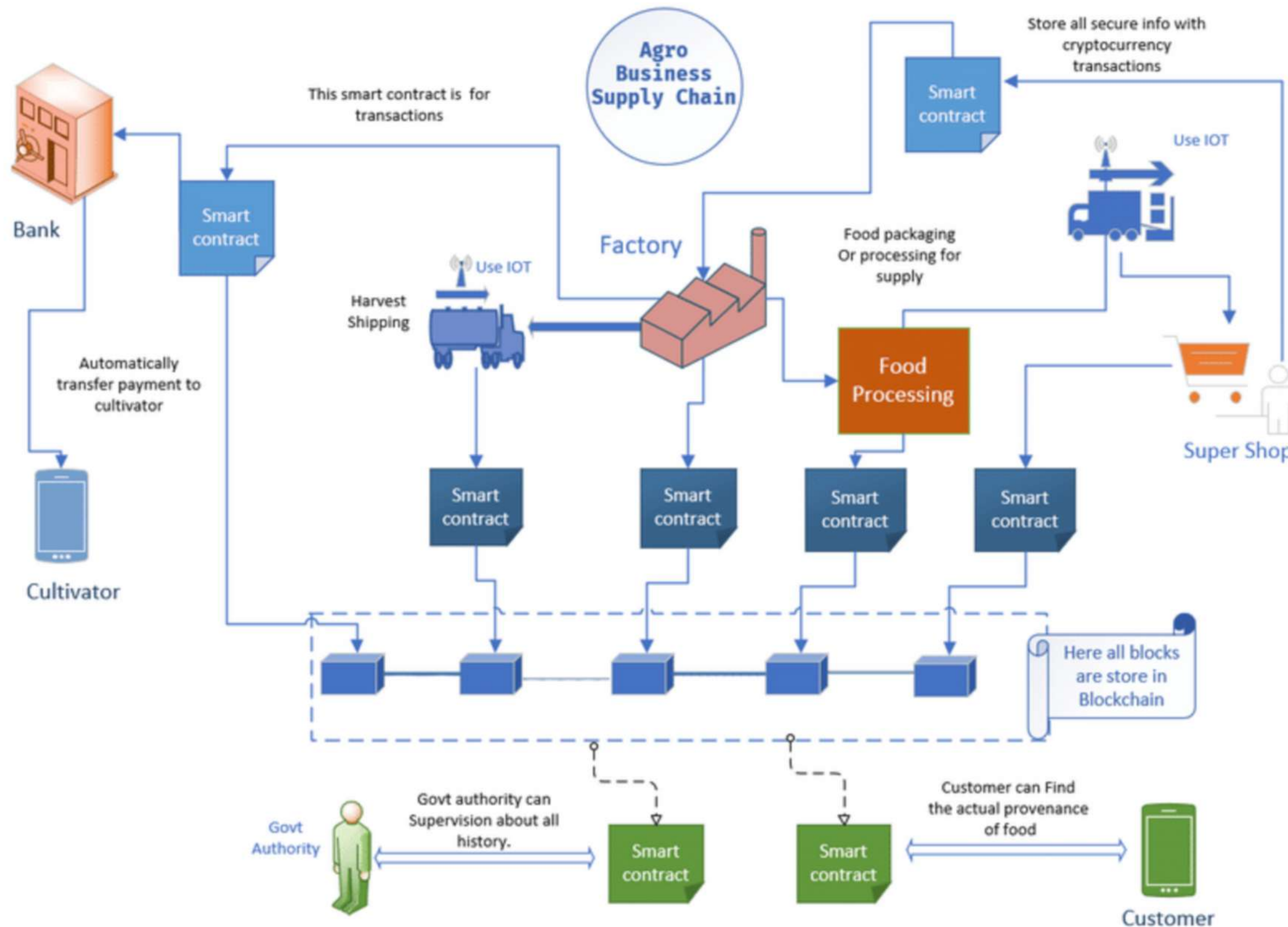
Blockchain enhances **traceability and transparency** in agricultural supply chains. By recording every transaction securely, it minimizes waste and inefficiencies, which directly contribute to emissions. For instance, it allows for better tracking of produce from farm to market, reducing logistical emissions and ensuring compliance with sustainability standards.

Energy-Efficient Machinery

Transitioning to **fuel-efficient machinery** or equipment that uses alternative fuels, such as electric or biodiesel-powered tractors, reduces emissions during farming operations.



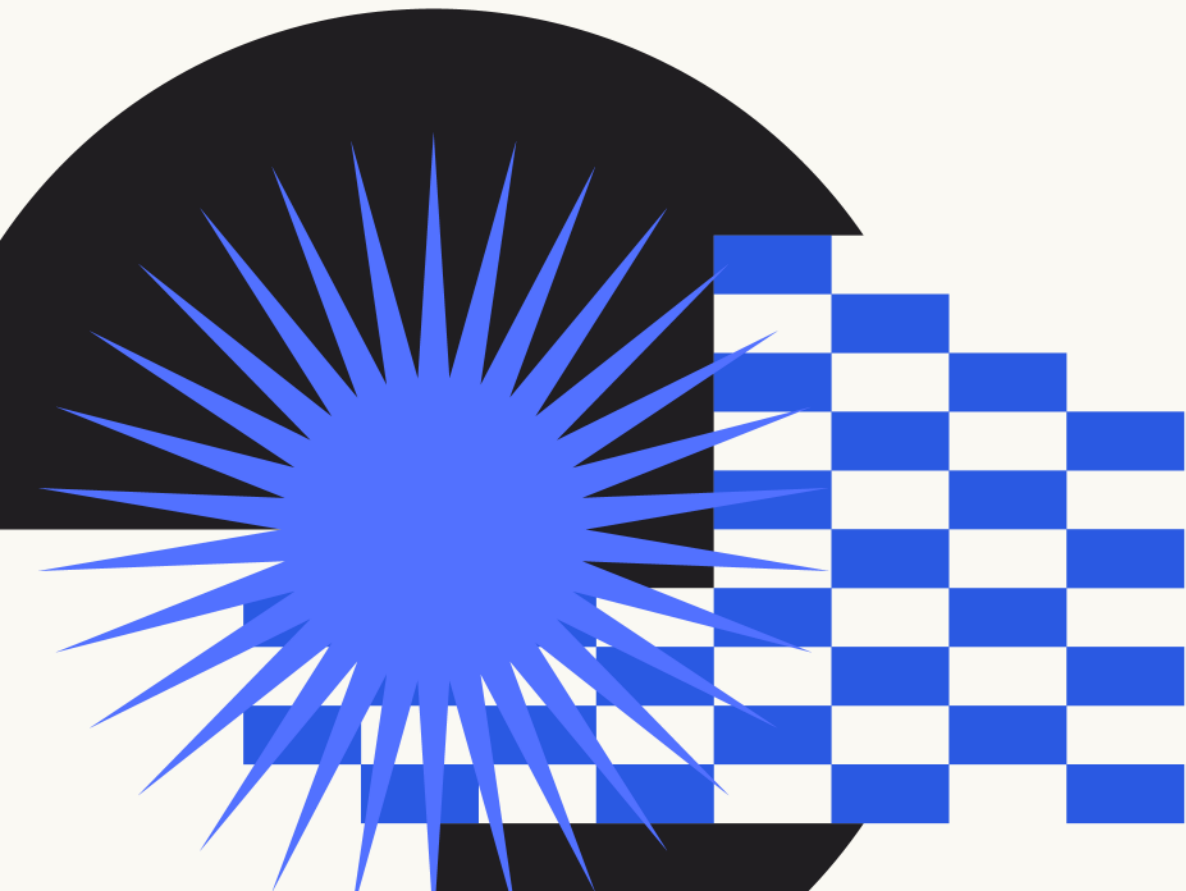
Supply Chain Workflow Diagram



The integration of digital twin and blockchain technologies in agriculture presents a significant opportunity for emission reductions. By leveraging data-driven insights and improving supply chain transparency, stakeholders can achieve substantial progress toward sustainability targets.

Balancing Carbon Emissions and Profitability

(Δ = Carbon Emissions - Profitability)



Concept Overview

The key challenge in sustainable supply chain management is to minimize the gap (Δ) between carbon emissions and profitability. The target is to reduce emissions while maintaining or increasing profitability.

Strategies to Minimize Δ

- **Adopting Renewable Energy Sources**
 - Impact on Emissions: Transitioning to solar or wind energy reduces carbon emissions significantly.
 - Profitability: Although the initial investment may be high, operational costs decrease over time, improving long-term profitability.
- **Precision Agriculture**
 - Impact on Emissions: Uses technology to optimize resource use, reducing fertilizer and pesticide overuse, and thus emissions.
 - Profitability: Enhanced yield and reduced input costs increase overall profitability.
- **Optimizing Logistics and Transportation**
 - Impact on Emissions: Route optimization and adopting fuel-efficient vehicles lower transportation emissions.
 - Profitability: Reduced fuel consumption leads to lower costs, directly benefiting the profit margins.
- **Circular Economy Approaches**
 - Impact on Emissions: Emphasizes recycling and waste reduction, leading to a smaller carbon footprint.
 - Profitability: Creates new revenue streams from recycled products, lowering waste disposal costs.

Balancing Carbon Emissions and Profitability

✂ Quantitative Assessment of Δ

- Δ = Carbon Emissions - Profitability Ratio: By quantifying emissions and profitability metrics (e.g., Return on Investment or ROI), businesses can track the effectiveness of emission-reducing strategies.
- Benchmarking: Establishing benchmarks for emission levels relative to profitability for different supply chain phases enables continuous performance tracking.

✂ Case Example: Agriculture in India

- Transitioning to drip irrigation reduces water usage by up to 40% and decreases emissions associated with pump electricity usage.
- Implementing smart sensors for soil monitoring not only cuts down emissions from unnecessary resource application but also boosts productivity by up to 30%, improving profitability.

✂ Key Considerations for Minimizing Δ

- Initial Investment vs. Long-term Gains: Emphasize strategies that might have a higher upfront cost but promise sustainable profitability.
- Scalability: Ensure solutions are scalable across regions and adaptable to different supply chain segments.
- Risk Management: Account for uncertainties in the market, such as fluctuations in energy prices or changes in environmental regulations.



Prediction of Carbon Emission Changes with Digital Twin and Blockchain in Agriculture



Case Study: Implementation in Indian Agriculture

- **Digital Twin in Smart Irrigation Management:**

- In Punjab, the use of **digital twin** technology for smart irrigation resulted in a **22% reduction** in water usage. This change led to a **15% decrease in fuel consumption** by irrigation equipment, reducing carbon emissions by approximately **8,000 tons CO2e** in the first year.

- **Blockchain in Supply Chain Logistics:**

- In Maharashtra, **blockchain** technology was integrated into the agricultural supply chain, focusing on logistics and distribution. By improving traceability and optimizing transportation routes, the implementation led to a **25% decrease in transport-related emissions**, which equates to about **12,000 tons CO2e reduction annually**.

Prediction with Data Analysis for the Next 5 Years

- **Forecasting Emission Reductions:**

- Digital Twin: Adoption of digital twin technology is expected to grow at a compound annual growth rate (CAGR) of 18% in Indian agriculture, driven by increasing investments in smart farming practices. This growth could lead to a cumulative 30-40% reduction in carbon emissions from on-farm activities over the next five years.
- Blockchain: With blockchain technology's integration in logistics expected to increase at a **22% CAGR**, the technology could contribute to a 10-20% reduction in supply chain emissions, equating to a potential reduction of **90-120 million tons CO2e by 2027**.

- **Quantitative Impact:**

- Digital Twins: If adopted widely, the technology could save approximately **50 million tons CO2e** annually by 2027 through improved resource management and operational efficiencies.
- Blockchain Integration: Blockchain could prevent roughly **30 million tons CO2e annually** in emissions by 2027 by reducing logistical inefficiencies and enhancing supply chain transparency.

Implementation Strategy

Phased Approach to Emission Reduction

- Phase 1: Pilot Testing - Begin with small-scale implementation of digital twins in key farming operations and blockchain in select logistics processes to identify inefficiencies.
- Phase 2: Scaling Up - Expand the technologies across broader supply chain activities, including distribution and retail, based on pilot results.

Stakeholder Collaboration:

- Engage with farmers, suppliers, logistics providers, and tech partners to align goals, share data, and co-create solutions.

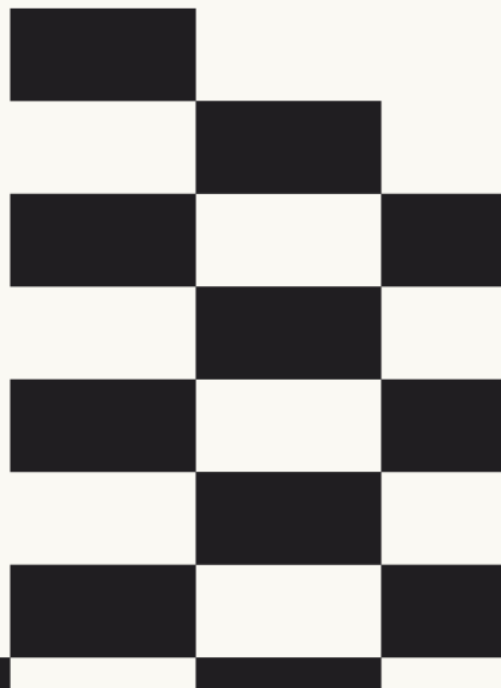
Technology Adoption Roadmap:

- Digital Twin Deployment: Utilize predictive analytics to simulate supply chain scenarios, optimize resource usage, and anticipate disruptions.
- Blockchain Integration: Implement traceable data systems to improve transparency, reduce fraud, and streamline logistics.

Performance Metrics and KPIs:

- Establish key performance indicators (KPIs) to measure carbon reduction, cost savings, and efficiency improvements from the use of these technologies.

This structured approach ensures a strategic, data-driven pathway to minimizing carbon emissions while maximizing supply chain efficiency.





→ Impact Assessment & Risk Analysis

Impact Assessment

Digital Twin Technology:

- **Positive Impact:** Digital twins enable real-time optimization, resulting in reduced carbon emissions, efficient resource utilization, and minimized waste. They provide insights for improving logistics and mitigating disruptions, which lead to a projected **20-30% reduction** in carbon emissions across supply chains.
- **Scalability and Efficiency:** Enhances decision-making by predicting supply chain performance and optimizing resource allocation, positively influencing sustainability goals.

Blockchain Technology:

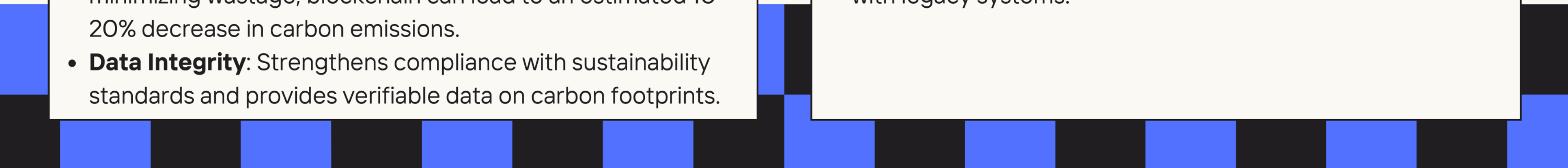
- **Positive Impact:** Blockchain's traceability ensures transparency in supply chains, reducing inefficiencies and preventing fraudulent practices. By improving logistics and minimizing wastage, blockchain can lead to an estimated 10-20% decrease in carbon emissions.
- **Data Integrity:** Strengthens compliance with sustainability standards and provides verifiable data on carbon footprints.

Risk Analysis

Digital Twin Risks:

- **High Implementation Cost:** The initial setup and integration of digital twins require significant investment in technology and skilled personnel, which can be a barrier for small-scale farms.
- **Data Security Concerns:** Handling real-time data could expose vulnerabilities to cyber-attacks, risking the integrity and confidentiality of the supply chain information.

Blockchain Risks:

- **Data Privacy:** Although blockchain is secure, concerns around data privacy may arise if sensitive information is shared among multiple stakeholders.
 - **Interoperability Issues:** Integrating blockchain technology into existing systems can be challenging, requiring significant changes to infrastructure and potential compatibility issues with legacy systems.
- 

Conclusions and Recommendations

NOW!

Conclusions

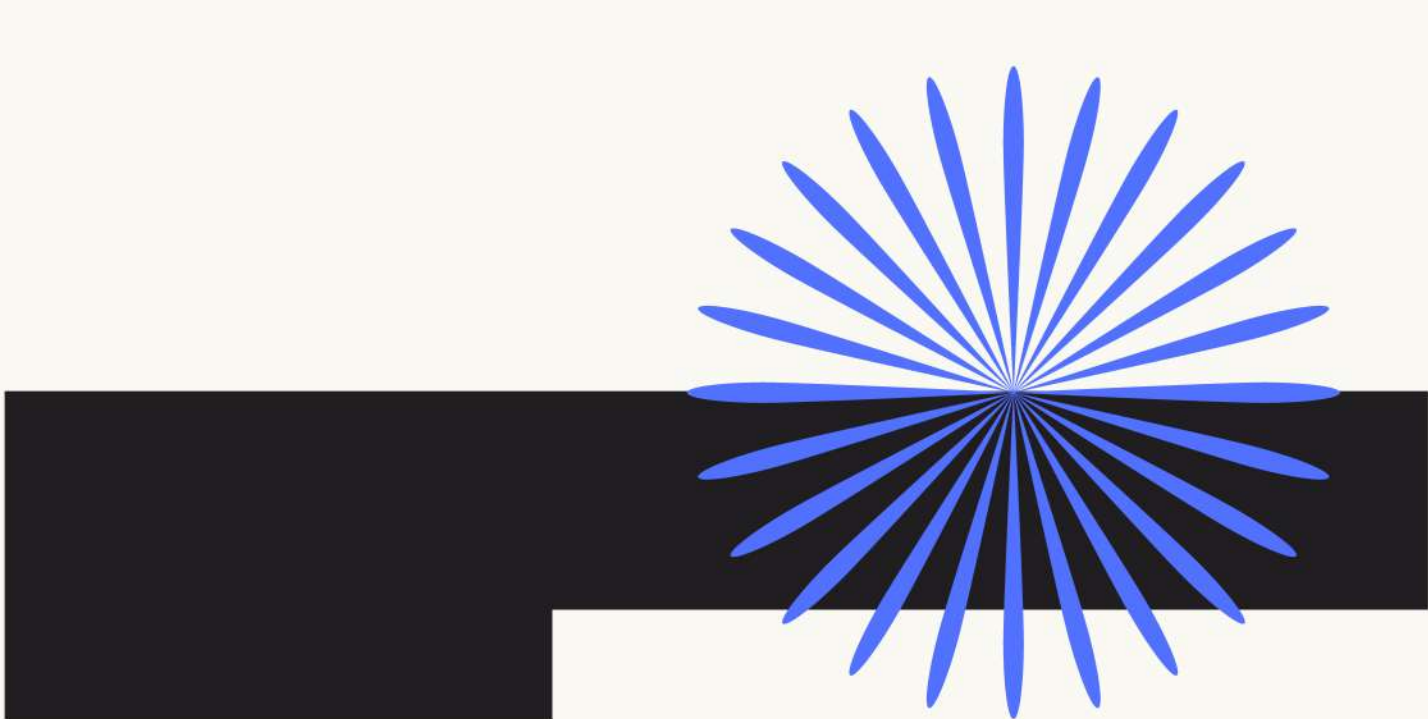
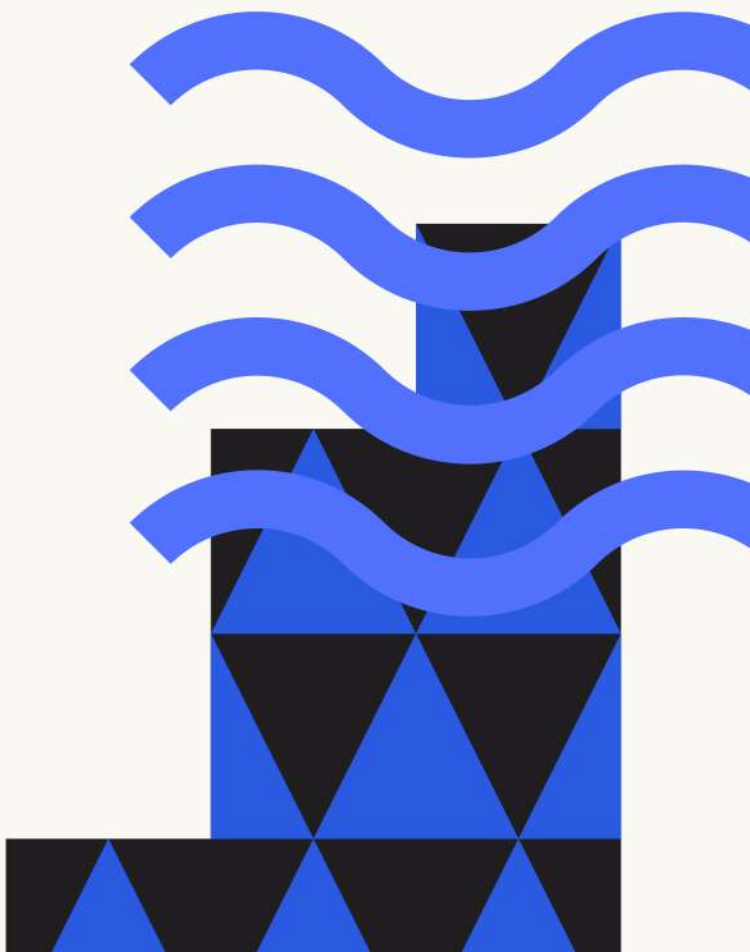
- **Significant Emission Reduction Potential:** The integration of digital twin and blockchain technologies in agricultural supply chains offers substantial potential for reducing carbon emissions while enhancing supply chain efficiency.
- **Enhanced Decision-Making:** Digital twins provide a detailed analysis of supply chain behavior, allowing for more informed decisions, while blockchain ensures transparency and accountability.
- **Challenges:** Despite the benefits, challenges such as high implementation costs, data security, and system integration need to be addressed for widespread adoption.

Recommendations

1. **Phased Implementation:** Begin with pilot projects to test and refine the technologies before scaling them to larger operations.
2. **Stakeholder Engagement:** Involve all stakeholders early to align goals, manage expectations, and ensure smooth adoption of new technologies.
3. **Policy Support:** Advocate for supportive policies and incentives from governments to encourage the adoption of digital and blockchain technologies in agriculture.
4. **Continuous Monitoring and Evaluation:** Establish robust frameworks to monitor and assess the impact of these technologies on carbon emissions and supply chain performance.
5. **Focus on Training and Capacity Building:** Invest in training programs to upskill the workforce to effectively manage and optimize these advanced technological systems.

This approach ensures a balanced focus on sustainability, profitability, and technological adoption in the agricultural sector.

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Thanks

Akelahu

Nikunj Sharma